

Unit VIII: Atoms and Nuclei

CHAPTER-12: ATOMS

GIST OF CHAPTER

Alpha-particle scattering experiment; Rutherford's model of atom; Bohr model, energy levels, hydrogen spectrum.

DEFINITIONS AND CONCEPT:

Atomic models

As atom is electrically neutral the discovery of electron led by J.J. Thomson established that it should also have positive charge. Hence, he proposed first model of atom- Plum- Pudding model.

Plum – Pudding Model

According to plum pudding model,” the positive charge of the atom is uniformly distributed throughout the volume of the atom and the negatively charged electrons are embedded in it like seeds in a watermelon.”

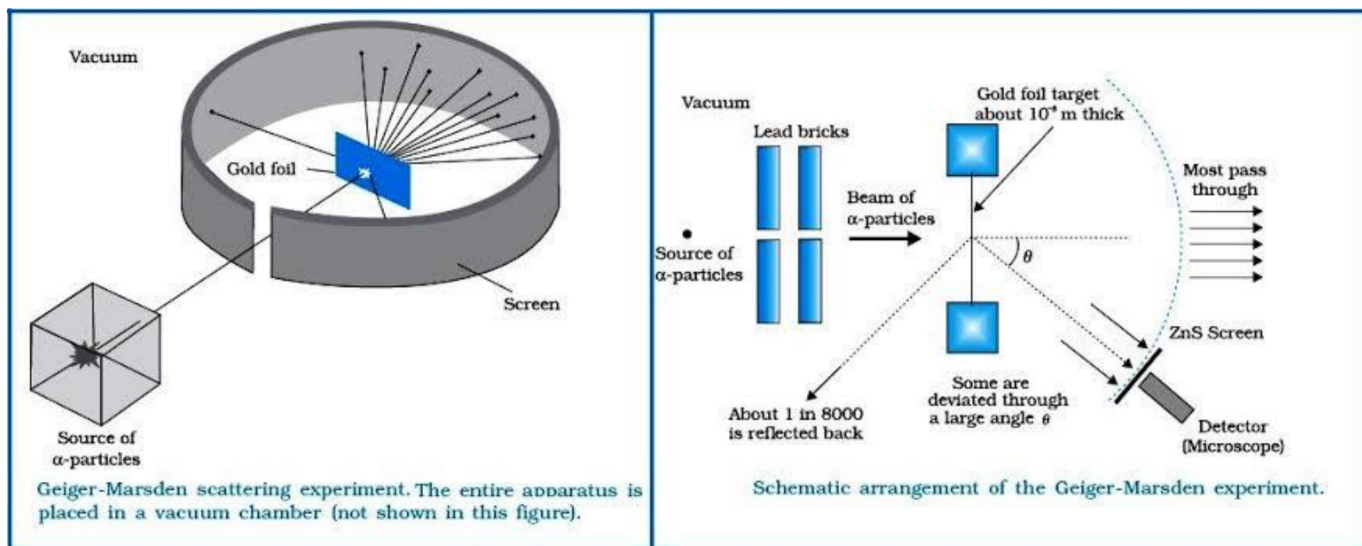
But subsequent studies on atom showed the results very different from this atomic model.

Rutherford’s atomic model

According to this model “ The entire positive charge and most of the mass of an atom is concentrated in a small volume called the nucleus, with electrons revolving around the nucleus.”

Geiger and Marsden experimentally proved Rutherford’s atomic model.

Alpha particle scattering Experiment



- Radioactive element ${}^{214}_{83}\text{Bi}$ was taken as α -particles generating source
- Gold was taken as target metal. The selection of gold was based upon its two important characteristics:

- a) Gold has the highest malleability. Gold foil that was used in the experiment was almost transparent.
- b) Gold is a heavy metal, hence it helped in discovery of the nucleus.
- Lead bricks absorbed the α -particles which were not in the direction of gold foil. They worked as collimator.
- The detector was made from ZnS.

Experimental Observations:

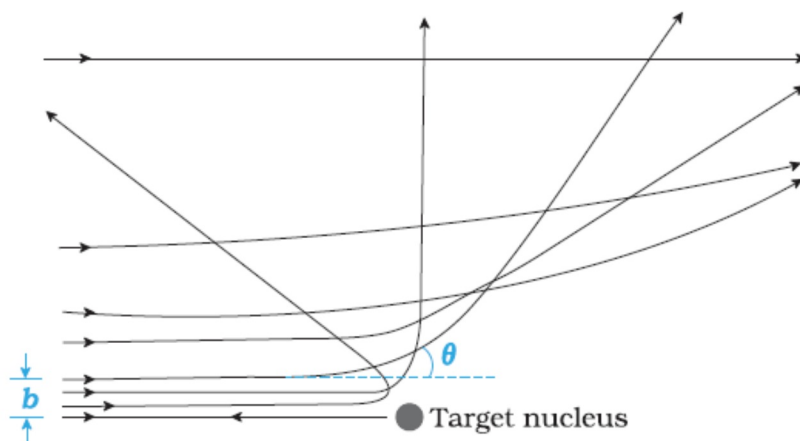
- When α -particles hit ZnS screen, it absorbs and glows. Hence the number of α -particles can be counted by intensity variation.
- Most of the α -particles passed roughly in a straight line (within 1°) without deviation. This showed that no force was acting upon most of the α -particles.
- A very small number of α -particles were deflected. (1 out of 8000)

Conclusions:

- Most of the space in the atom is mostly empty (only 0.14% scatters more than 1°)
- Experiment suggest that all positively charged particles are together at one location at centre. It was called nucleus. So, nucleus has all the positive charges and the mass. Therefore, it has the capability to reflect heavy positive α -particles.
- Size of nucleus is calculated to be about 10^{-15}m to 10^{-14}m . According to kinetic theory, size of atom is of the order of 10^{-10}m .
- Force between α -particles and gold nucleus $F = \frac{1}{4\pi\epsilon_0} \frac{(2e)(Ze)}{r^2}$

Alpha-particle trajectory:

- Impact parameter: It is the perpendicular distance between the directions of given α -particle and the centre of the nucleus. It is represented by 'b'.



- Distance of closest approach: It is the distance between centre of nucleus and the α -particle when it stops and reflects back. It is represented by 'd'. This distance gives an approximation of nucleus size.

Electron Orbits:

- We can calculate the energy of an electron and the radius of its orbit based upon Rutherford model.
- The electrostatic force of attraction, F_e between revolving electron and the nucleus provides the requisite centripetal force (F_c) to keep them in their orbits.

$$F_e = F_c$$

$$\text{For hydrogen atom, } \frac{1}{4\pi\epsilon_0} \frac{e^2}{r^2} = \frac{mv^2}{r}$$

$$r = \frac{e^2}{4\pi\epsilon_0 mv^2}$$

Electron has kinetic energy, $K = \frac{1}{2}mv^2$. Putting the value of mv^2 in the above equation

$$K = \frac{e^2}{8\pi\epsilon_0 r}$$

$$\text{And } v = \frac{e}{\sqrt{4\pi\epsilon_0 mr}}$$

P.E. of an electron, $U = -\frac{1}{4\pi\epsilon_0} \frac{e^2}{r}$ negative sign shows that it is due to attractive force.

Total Energy, $E = K + U$

$$\begin{aligned} E &= \frac{e^2}{8\pi\epsilon_0 r} + \left[-\frac{1}{4\pi\epsilon_0} \frac{e^2}{r} \right] \\ &= -\frac{e^2}{8\pi\epsilon_0 r} \end{aligned}$$

- Due to this negative energy, the electron is bound to the nucleus and revolves around it. This energy is known as the binding energy of an electron.
- From the equation, it is clear that if energy is zero, then radius is infinity. Practically, if we provide this amount of energy to this electron, it gets free.

Atomic Spectra:

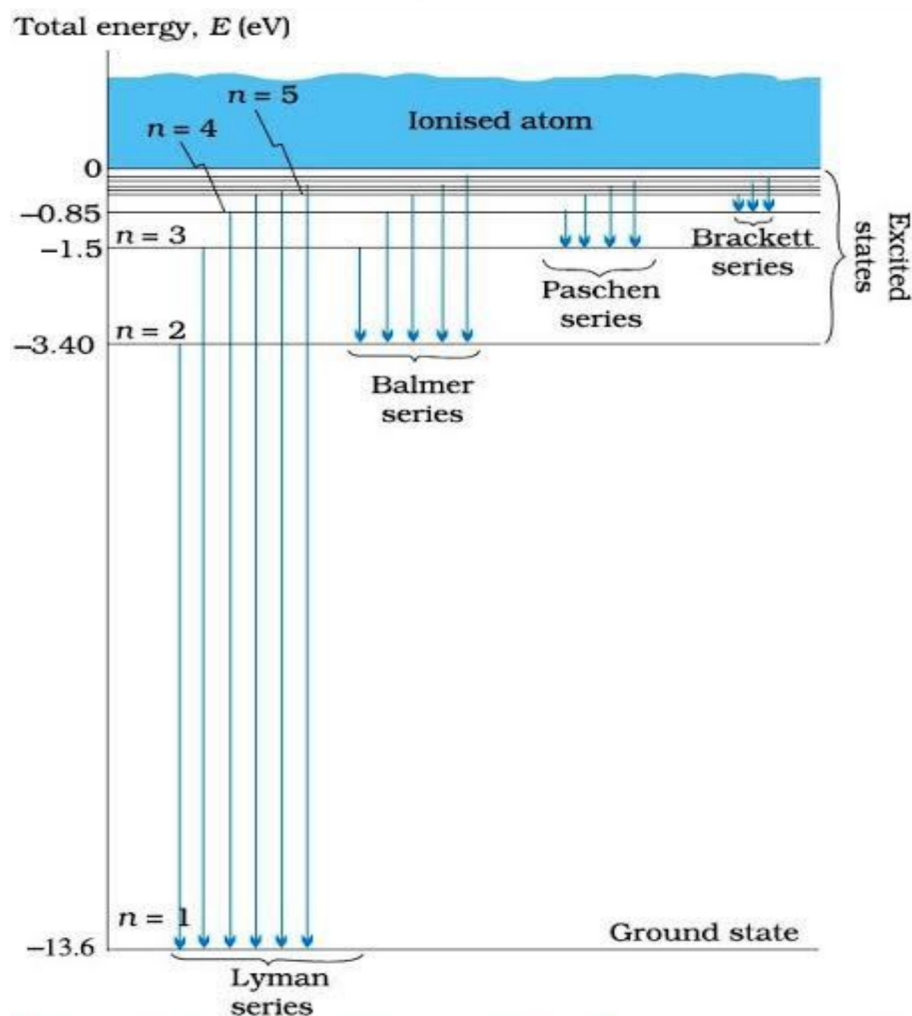
- Each element has a characteristics spectrum of radiation, which it emits. There are two types of atomic spectra: Emission atomic spectra and absorption atomic spectra.
- Emission atomic spectra: Due to excitation of atom usually by electricity, light of particular wavelength is emitted. This atomic spectra is known as emission spectra.
- Absorption atomic spectra: If atoms are excited in presence of white light, it absorbs it absorbs its emission spectral colours and black lines appear in the same places of that atoms emission spectra. This type of spectra is known as absorption spectra.

Spectral Series:

- The atom shows range of spectral lines. Hydrogen is the simplest atom and has the simplest spectrum.
- The spacing between lines within certain sets of the hydrogen spectrum decreases in a regular way. Each of these sets is called a spectral series.
- Balmer Series: Balmer observed the first hydrogen spectral series in visible range of the hydrogen spectrum. It is known as Balmer Series.

$$\frac{1}{\lambda} = R \left[\frac{1}{2^2} - \frac{1}{n^2} \right] \text{ where } n=3,4,5\dots \text{ and } R \text{ is Rydberg's constant.}$$

The value of R is $1.097 \times 10^7 \text{ m}^{-1}$;



Line spectra originate in transitions between energy levels.

Other series of spectra for hydrogen were as follows

- Lyman Series : $\frac{1}{\lambda} = R \left[\frac{1}{1^2} - \frac{1}{n^2} \right] \text{ } n=2,3,4,5\dots$ This is in UV range

- Paschen Series: $\frac{1}{\lambda} = R \left[\frac{1}{3^2} - \frac{1}{n^2} \right]$ $n=4,5,6,\dots$ This is in IR range
- Brackett Series: $\frac{1}{\lambda} = R \left[\frac{1}{4^2} - \frac{1}{n^2} \right]$ $n=5,6,7,\dots$ This is in IR range
- Pfund Series: $\frac{1}{\lambda} = R \left[\frac{1}{5^2} - \frac{1}{n^2} \right]$ $n=6,7,8,\dots$ This is in IR range

Limitations of Rutherford model:

- **It could not explain the stability of the atom:** The electron orbiting around the nucleus radiates energy. As a result, the radius of electron orbit should continuously decrease and ultimately the electron should fall into the nucleus.
- **It could not explain the nature of energy spectrum:** According to the Rutherford's model, the electron can revolve around the nucleus in all possible orbits. Hence, the atom should emit radiations of all possible wavelengths or in other words, it should have a continuous spectrum. However, in practice, the atoms are found to have a line spectrum or discrete spectrum.

BOHR'S MODEL AND POSTULATES:

- An electron can revolve in certain stable orbits without emission of radiant energy. These orbits are called stationary states of the atom.
 - Electron revolves around nucleus only in those orbits for which the angular momentum is the integral multiple of $\frac{h}{2\pi}$, where, h is Planck's constant.
- Hence, angular momentum, $L = \frac{nh}{2\pi}$
- An electron may make a transition from one of its specified non-radiating orbit to another of lower energy. When it does so, a photon of energy $h\nu$ is radiated having energy equal to energy difference between initial and final state.

$$h\nu = E_i - E_f \text{ (where, } \nu \text{ is frequency)}$$

$$\text{Angular momentum, } L = mv_n r_n = \frac{nh}{2\pi} \Rightarrow mr_n = \frac{nh}{2\pi v_n}$$

$$\text{For hydrogen atom, orbital velocity of electron } v = \frac{e}{\sqrt{4\pi\epsilon_0 m r}}$$

$$\text{Combining these two equations, we get } v_n = \frac{1}{n} \frac{e^2}{4\pi\epsilon_0} \frac{1}{h/2\pi}$$

This equation depicts that electron speed in n^{th} orbit falls by a n factor

$$r_n = \left[\frac{n^2}{m} \right] \left[\frac{h^2}{2\pi^2} \right] \left[\frac{4\pi\epsilon_0}{e^2} \right]$$

For innermost orbit $n=1$; value of r_1 is known as Bohr's radius a_0

$$a_0 = \frac{h^2 \epsilon_0}{\pi m e^2}$$

If we put the values of all constants, we get **$a_0 = 5.29 \times 10^{-11} \text{ m}$**

It can also be observed that radii of n^{th} orbit increases by n^2 times

By putting this value in total energy of electron and convert the unit in eV, we get

$$E_n = \frac{-13.6}{n^2} \text{ eV. Negative value shows that electron is bound to nucleus.}$$

De-Broglie's explanation of Bohr's second postulate by quantization theory:

- According to Bohr's postulate, an electron in a hydrogen atom can revolve in a certain orbit only in which its angular momentum, $L = \frac{nh}{2\pi}$. In these stationary orbits, an electron does not radiate energy.
- De-Broglie proved it with the help of the wave nature of electron.
- Travelling wave propagates energy but stationary wave does not propagate energy. In analogy to waves travelling on a string, particle waves can lead to standing waves under resonant conditions. Resonant condition is $l = 2\pi r$ where, $l =$ perimeter of orbit.

For a hydrogen atom, length of the innermost orbit is its perimeter.

$$\text{Hence } 2\pi r = n\lambda \quad \dots\dots(i)$$

According to de-Broglie's wavelength of electron,

$$\lambda = \frac{h}{p}$$

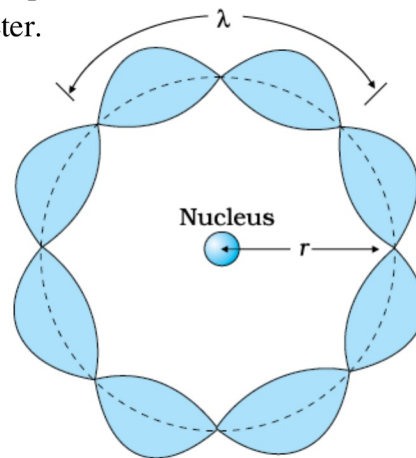
Here $p = mv$, Now equation (i) can be written as

$$2\pi r = n \frac{h}{p} = n \frac{h}{mv} \quad \dots\dots(ii)$$

Hence, equation (ii) can be reduced as,

$$mvr = n \frac{h}{2\pi} \Rightarrow L = \frac{nh}{2\pi}$$

This is Bohr's second postulate.



Limitation of Bohr's atomic model:

Bohr's model is for hydrogenic atoms. It does not hold true for a multi-electron model.

Formulae:-

- Radius of orbit $r = \frac{e^2}{4\pi\epsilon_0 mv^2}$
- Kinetic energy of electron in its orbit $K = \frac{e^2}{4\pi\epsilon_0 r}$
- Potential energy of an electron $U = -\frac{e^2}{4\pi\epsilon_0 r}$
- Velocity of electron in its orbit $v = \frac{e}{\sqrt{4\pi\epsilon_0 mr}}$
- Total energy of an electron in an orbit $E = \frac{e^2}{8\pi\epsilon_0 r}$
- Balmer Series : $\frac{1}{\lambda} = R \left[\frac{1}{2^2} - \frac{1}{n^2} \right]$ $n=3,4,5\dots$ This is in Visible range
- Lyman Series : $\frac{1}{\lambda} = R \left[\frac{1}{1^2} - \frac{1}{n^2} \right]$ $n=2,3,4,5\dots$ This is in UV range
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- Relation between speed, total energy of an electron and its radius with respect to orbital

number n: Speed of electron $v_n = \frac{1}{n} \frac{e^2}{4\pi\epsilon_0} \frac{1}{h/2\pi}$

Radius of orbit $r_n = \frac{n^2}{m} \frac{h}{2\pi} \frac{4\pi\epsilon_0}{e^2}$

Bohr radius, $a_0 = \frac{h^2 \epsilon_0}{\pi m e^2} = 0.53 \text{ \AA}$

Energy for n^{th} orbit electron, $E_n = \frac{-13.6}{n^2} \text{ eV}$

MNEMONICS for Concept: Hydrogen Spectra

Mnemonics: Papa	Brings	Pastry for	Babu and	Lal
Pfund	Brackett	Paschen	Balmer	Lyman
$n_1=5$	$n_1=4$	$n_1=3$	$n_1=2$	$n_1=1$

MNEMONICS for Concept: Range of each series of Hydrogen Spectra

1 is Unimportant, **2** is Very important, **Rest** are Important
 $n_1=1$ UV range $n_1=2$ Visible range $n_1=3,4,5$ IR range